REPUBLIC OF KENYA

MINISTRY OF SPORTS

TALANTA STADIUM SMART INFRASTRUCTURE SYSTEM

Integrated Digital Ticketing | Biometric Access Control
Venue & Workforce Management | Real-time Operations

— "A Smarter Stadium. A Safer Nation." —

DOCUMENT TYPE:	Government-Focused Technical & Financial Proposal
PROPOSING ENTITY:	ST-Tech ICT Consultants
TARGET VENUE:	Talanta Sports City / Talanta Stadium (60,000 Capacity), Nairobi, Kenya
PROPOSED CLIENT:	Ministry of Sports / Relevant Government Stadium Operating Authority
INDICATIVE BUDGET:	KES 75,000,000 preliminary planning envelope (subject to scope validation)
TIMELINE:	Indicative 10-month delivery roadmap post-contract execution
REFERENCE BASIS:	Presidential Directive / Public Procurement Framework Context

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DOCUMENT NAVIGATION GUIDANCE:
This proposal is organized into 35 numbered sections across a 38-page structure. Pages 4 through 28 deliver the core technical and financial proposal, while Pages 29 through 38 contain the Technical Annex with buildable engineering blueprints, API contracts, data lifecycles, and procurement matrices.

1. Proposal Overview and Commercial Basis

Target Venue: Talanta Sports City / Talanta Stadium (60,000-Seat Capacity), Nairobi, Kenya.

Proposed Client: Ministry of Sports / Relevant Government Stadium Operating Authority.

Proposing Firm: ST-Tech ICT Consultants.

Proposal Status: Expanded Concept Proposal for Evaluation and Procurement Planning.

Commercial Basis: Indicative Planning Envelope (Preliminary; Non-Binding Quotation).

Reference Directive: Presidential Directive / Public Procurement Framework Context.

This document presents a comprehensive, government-aligned technical and commercial proposal for an Integrated Smart Infrastructure System at Talanta Stadium. The proposed solution establishes an end-to-end digital operating framework encompassing public e-ticketing, secure payment gateways, gate validation, workforce accreditation, real-time command dashboards, and financial reconciliation.

> ***IMPORTANT NOTICE:*** *Prepared as an expanded concept proposal. No formal government endorsement, contract award, construction partnership, or live system deployment is implied. All technical scope, financial figures, integration interfaces, and performance statements remain subject to formal site survey, client verification, and public procurement procedures.*

2. Document Control, Purpose and How to Use This Proposal

This document describes a proposed digital platform that could be developed if ST-Tech ICT Consultants is selected through the applicable public procurement process. It supports stakeholder discussion and tender planning—not substituting for a formal tender response, signed contract, detailed design, or statutory approvals.

Field Metadata Breakdown:

- Document Title: Integrated Talanta Stadium Smart Digital Infrastructure Technical & Financial Proposal
- Version Reference: Expanded Concept Proposal – Final Polish & Review Draft
- Venue Context: Talanta Sports City / Talanta Stadium, Nairobi, Kenya (60,000 Capacity)
- Proposer: ST-Tech ICT Consultants (Authorised Representative & Technical Lead)
- Planning Envelope: KES 75,000,000 preliminary planning envelope carried forward from initial draft
- Implementation Period: Indicative 10-month delivery roadmap post-contract execution and site access
- Required Approvals: Client ICT, Venue Operations, Ministry of Finance, Office of the Data Protection Commissioner (ODPC), Communications Authority (CA), and Procurement Authorities

> ***IMPORTANT NOTICE:*** *All visual mockups, architectural diagrams, and flowcharts presented in this document are illustrative concepts created to communicate system layout and functional relationships. They do not represent installed hardware, active software instances, or pre-approved branding.*

3. Executive Summary

Talanta Stadium is a flagship national infrastructure project designed to anchor major regional and international sporting events. The opportunity proposed by ST-Tech ICT Consultants is to build a unified digital operating layer surrounding event discovery, ticket issuance, cashless payment processing, gate entry validation, staff accreditation, matchday command operations, and automated financial settlement.

The proposed Integrated Talanta Stadium Digital Infrastructure (ITSDI) utilizes a modular, micro-service-ready architecture. This enables the venue operator to deploy core ticketing, payment, and gate validation systems first, followed by incremental integration of advanced workforce, accreditation, and analytical modules once operational baseline stability is verified.

Core Proposed Outcomes:

- Official Public Channel: Unified mobile-first web portal for fixture discovery, ticket purchasing, matchday instructions, and customer service.
- Financial Integrity & Settlement: Real-time M-PESA and card transaction verification, automated refund handling, and event-level revenue reconciliation.
- High-Throughput Gate Control: Rapid ticket validation workflows backed by offline cached contingency rules and supervisor exception handling.
- Workforce & Accreditation Management: Centralised role-based credentialing, shift scheduling, vendor attendance tracking, and zone access control.
- Unified Command Centre: Real-time operational dashboards providing attendance counts, gate health metrics, queue warnings, and incident tracking.
- Privacy & Security by Design: Strict compliance with Kenya's Data Protection Act 2019, TLS/AES encryption, MFA for administration, and full auditability.

Formal Decision Sought: Approve a formal Discovery & Feasibility Phase to validate venue site readiness, existing infrastructure interfaces, network cabling, power distribution, data protection parameters, and public procurement requirements prior to finalizing a binding contract.

4. Public-Sector Case and Strategic Alignment

A government-grade digital infrastructure proposal must deliver tangible public value, financial accountability, and operational efficiency. The ITSDI platform directly aligns with national digital transformation priorities and stadium management objectives.

> ***IMPORTANT NOTICE:*** *System performance improvements and risk reductions must be measured against verified venue baseline metrics established during the Discovery Phase.*

Public Sector Objective	Proposed System Contribution	Target Evidence / KPI
Financial Accountability	Traceable transaction logs, automated M-PESA reconciliation, audit-ready revenue reports per event.	Signed event reconciliation statement within 24 hours of match closeout.
Safer Event Flow	Live gate validation rates, density heatmaps, queue bottleneck alerts, and structured escalation.	Median gate throughput <2.5 seconds; zero unhandled turnstile queues.
Convenient Public Service	Mobile-optimised booking, instant digital ticket delivery, self-service recovery, accessible seating.	>90% checkout completion rate; customer support response <5 mins.
Operational Efficiency	Automated roster scheduling, zone-specific accreditation, shift timekeeping, vendor invoice verification.	<2% variance between scheduled staff shifts and vendor billing claims.
Local Capability Transfer	Comprehensive technical training, operational runbooks, administrator handover, ongoing local IT support.	100% pass rate on operator competency evaluations prior to go-live.
Privacy & Digital Trust	Data minimisation, clear privacy notices, role-based access restrictions, strict retention schedules.	Zero non-compliant data disclosures; completed DPIA assessment.

5. Venue Context, Construction Background and Boundaries

Publicly available reporting confirms that China Road and Bridge Corporation (CRBC) is the main engineering contractor for Talanta Sports City, with technical supervision and project execution oversight provided by the Kenya Defence Forces (KDF) Engineering Department. Architectural and design consultancies have been associated with University of Nairobi Enterprises and Services Limited (UNES) under Prof. Erastus Abonyo.

Construction officially commenced in March 2024. A CRBC project update dated 10 February 2026 targeted substantial civil completion by late Q1 2026. The actual current status of structural handover, power connection, and site readiness must be formally confirmed with venue authorities before physical installation begins.

Digital Project Boundaries:

- Core Focus: Software platform, API integration layer, mobile scanner applications, data management, and operational dashboards.
- Physical Hardware: Turnstile barriers, CCTV cameras, servers, network cabling, access gates, and power systems require independent procurement or integration with existing venue contracts.
- Third-Party Approval: Integrations with external payment providers (Safaricom M-PESA) or government identity registers require formal legal authorization and technical interface documentation.

6. Scope of Work and Deliverables

The proposed implementation is structured into eight core workstreams, ensuring full coverage from requirements gathering to post-launch hypercare.

IMPORTANT NOTICE: *Turnstile barriers, optical cameras, network cabling, power UPS systems, SMS telecom charges, and payment gateway fees require separate commercial confirmation.*

Workstream	Included Scope & Technical Capabilities	Key Deliverable / Acceptance Target
1. Discovery & Design	Workshops, requirements, service maps, site survey, system inventory, risk and dependency analysis.	Approved requirements and solution design pack.
2. Public Ticketing	Fixture pages, ticket categories, booking, secure references, confirmation, support and refund workflows.	Tested web application and admin workflow.
3. Payments	Approved provider integration, callback handling, transaction states and reconciliation.	Sandbox evidence and production approval checklist.
4. Gate Validation	Ticket validation APIs, scanner workflows, exception queue and offline procedure.	Test report against approved gate devices.
5. Workforce	Staff/vendor records, roles, accreditation, roster, attendance and expiry alerts.	Working staff portal and exportable reports.
6. Command Dashboard	Event view, gate/device health, incident queue, attendance and revenue summaries.	Role-based dashboards and audit exports.
7. Security & Privacy	Threat model, privacy review, access controls, test support, retention and incident playbooks.	Review reports and remediation evidence.
8. Training & Handover	Admin/operator training, user guides, support model, recovery drill and documentation.	Signed training and handover pack.

7. Public Website and Mobile Experience

The public web and mobile portal serves as the primary digital touchpoint for stadium patrons, offering a secure, accessible, and fast-loading experience across mobile networks.

Core User Journeys:

- Home & Fixture Discovery: Official notices, upcoming match details, venue guidelines, and service status alerts.
- Event Selection & Booking: Interactive zone/seat selection, transparent pricing breakdowns, accessible seating options, and countdown hold timers.
- Cashless Checkout: Seamless M-PESA STK Push prompt, card payment options, instant transaction verification, and downloadable PDF/digital wallet tickets.
- Digital Ticket Wallet: Secure dynamic QR codes, offline ticket view capability, ticket transfer controls, and self-service support.
- Help & Customer Care: Frequently asked questions, refund policies, lost ticket recovery, accessibility assistance, and contact forms.

Design Standards: Target WCAG 2.2 Level AA accessibility compliance, mobile-first responsive design for 3G/4G networks, screen reader compatibility, and high-contrast visual modes.

8. Ticketing and Payment Workflow

To guarantee financial integrity and prevent fraudulent ticket generation, payment confirmation and ticket issuance are strictly decoupled and executed asynchronously.

> ***IMPORTANT NOTICE:*** *A user should never receive a confirmed ticket solely because a browser displayed a success screen; the backend must verify the provider callback or status before confirming the order.*

Order State	System Action & Behaviour	Control Safeguard
Draft Order	Hold selected seat/zone inventory for a defined configurable period.	Expiry and inventory locking rules; concurrency locks.
Payment Pending	Show pending state and reference; allow safe status refresh.	Idempotency and callback verification; polling limits.
Paid / Confirmed	Issue signed ticket reference, QR code, and e-receipt; notify user.	Unique order ID, cryptographic signature, audit record.
Failed / Timed Out	Explain next step; do not issue paid ticket until confirmed.	Re-query provider status; prevent duplicate charges.
Refund / Cancellation	Apply authorised rules, revoke ticket credentials, log status.	Role-based approval, two-factor authorization, audit log.

9. High-Level System Architecture

The ITSDI platform employs a modern multi-tier architecture designed for high availability, fast response times, and security.

Architectural Principles:

- Public Channel: Browser-based website, mobile web, and optional notifications.
- API Layer: Authentication, authorisation, rate limiting, input validation, and central audit events.
- Business Services: Ticket inventory, payments, ticket validation, accreditation, workforce, reports, and notifications.
- Data Layer: Relational transactional data, controlled document/object storage, audit log store, and backups.
- Venue Edge: Locally available validation data only where approved; synchronisation and documented offline rules.
- External Services: Payment provider and any government identity services only after formal approval and interface confirmation.

10. Detailed Architecture, Data Flow and Environments

System Tier Breakdown & Deployment Environments:

> ***IMPORTANT NOTICE:*** *Deployments are segregated into Development, Staging/Test, Production, and Training environments. Secrets must never be committed to source code.*

Layer	Responsibilities & Capabilities	Design Requirements & Safeguards
Experience Tier	Public site, staff portal, steward interface, admin console.	Responsive, accessible, role-aware, safe session handling.
Application / API Tier	Ticketing, inventory, payments, event configuration, accreditation, reporting.	Versioned APIs, validation, idempotency, audit hooks.
Integration Tier	Payment callbacks, scanners, SMS/email, approved external systems.	Timeouts, retries, provider status checks, secret management.
Persistence Tier	Orders, ticket state, event inventory, roles, logs and reports.	Backups, encryption, retention, and least privilege access.
Operations Tier	Monitoring, deployment, backups, alerting and incident response.	Separate environments, controlled releases, automated rollback.

11. Core Data Model and System Entities

A standardized relational data model prevents ticket duplication, inventory overselling, and financial discrepancies.

IMPORTANT NOTICE: *Do not collect a data field merely because it may be useful later. Define the purpose, lawful basis, access roles, retention period, deletion procedure, and breach response for each category.*

Entity Name	Key Attributes & Relationships	Essential Governance Controls
Event	Event ID, venue, date/time, status, entry rules, ticket categories.	Approved publishing; edit history audit log.
Ticket Inventory	Category, seat/zone, allocation, price, availability.	Concurrency control; no overselling under load.
Order / Payment	Order ID, amount, currency, provider reference, status, timestamps.	Idempotency; verified status; reconciliation.
Ticket	Ticket ID, order ID, event ID, status, validation events.	Single-use rules; revocation; anti-replay.
User / Role	Account ID, role, permissions, contact preferences.	MFA for privileged users; periodic access reviews.
Accreditation	Person/organisation reference, role, zone, validity period.	Purpose limitation; expiry; approval log.
Audit Event	Actor, action, object, timestamp, result, correlation ID.	Tamper-resistant write-once storage; restricted access.
Incident / Device	Gate/device identifier, status, event, assigned responder.	Severity, escalation and closure record.

12. Gate Access, Ticket Validation and Crowd Operations

Gate validation must be fast, observable and safe. The precise validation method depends on existing devices and the operator's security model.

Proposed Validation Sequence:

- Scanner reads the ticket token and verifies signature/status through the approved validation service.
- System checks event, entry window, ticket category, revocation and prior-use rules.
- Approved result returns clear accept/reject/exception status with minimal personal data displayed.
- A successful entry is recorded with gate/device, event, timestamp and ticket reference.
- An exception routes to a trained supervisor rather than forcing an unsafe decision.
- If network connectivity is lost, operators follow a rehearsed fallback with post-event reconciliation.

> ***IMPORTANT NOTICE:*** *Digital access control systems strictly serve venue operations and must NEVER impair physical emergency egress. All turnstiles and gates must incorporate hardware-level fail-safe mechanical overrides compliant with national safety standards.*

13. Identity Verification and Optional Biometrics

Biometric facial recognition is a sensitive, high-impact option—not a default requirement. It should be considered only if the stadium operator demonstrates a specific need that cannot be reasonably met by less intrusive methods, and after legal, privacy, accuracy, bias, accessibility and security review.

Required Decision Gates Before Deployment:

- Define precise purpose and lawful basis; obtain legal and data-protection review.
- Complete a Data Protection Impact Assessment (DPIA) and consult the relevant authority.
- Assess whether QR/NFC credentials or visual ID checks can achieve the purpose with less risk.
- Define enrollment, consent, retention, deletion, access, breach handling and vendor restrictions.
- Test false acceptance/rejection, demographic performance, lighting, and environmental conditions.
- Provide an accessible, staffed non-biometric alternative and appeal process for failed matches.
- Prohibit secondary use or surveillance expansion without a new approved assessment.

> ***IMPORTANT NOTICE:*** *No performance figures in this proposal should be treated as a vendor-certified biometric guarantee. Any accuracy or speed target must be specified, independently tested in representative conditions, and contractually defined before deployment.*

14. Staff, Contractor and Accreditation Management

The workforce module coordinates internal stadium staff, security personnel, contractors, catering vendors, media, and VIP guests.

IMPORTANT NOTICE: *The platform assists—does not unilaterally determine—employment, payroll, police clearance, or medical decisions.*

Capability	Operational Workflow	Administrative Control
Onboarding	Create record, organisation, role, approved documents and validity dates.	Authorised approver; minimum necessary data.
Shift Planning	Assign event, shift, supervisor and duty location.	Conflict checks and version history.
Accreditation	Issue credential with permitted zones and active time window.	Least privilege; expiry and revocation.
Attendance	Record check-in/out by approved method and flag exceptions.	Supervisor review; no automatic payroll approval.
Vendor Reporting	Compare approved rosters, attendance and submitted service records.	Finance approval and audit trail.
Offboarding	Expire credential and remove account access at end of assignment.	Revocation confirmation.

15. Command Centre, Dashboards and Reporting

A command-centre dashboard provides role-specific views rather than a single screen containing every sensitive detail. Priority is clear, timely operational information and actionable alerts.

IMPORTANT NOTICE: *Reporting principles require displaying timestamps, separating confirmed counts from estimates, and enforcing export controls for sensitive identity data.*

Dashboard View	Indicative Information Displayed	Authorised Users
Event Overview	Event status, scan counts, gate health, service alerts and incident queue.	Event control / venue management.
Gate Operations	Validation rates, rejects, device heartbeat and exception trends.	Gate supervisors / technical support.
Finance Console	Orders, paid/failed/pending, refunds, payment settlements and variances.	Authorised finance / audit.
Workforce Console	Shift coverage, attendance exceptions, accreditation expiry and vendor rosters.	Operations / HR / contract managers.
Security / Incident	Approved incident records, escalation status and response owner.	Authorised security personnel.
Technical Health	Application errors, latency, queue depth, storage, network and backup status.	ICT and support team.

16. Cybersecurity, Privacy and Compliance Plan

Security and data privacy are embedded into every layer of the platform architecture in compliance with Kenyan laws and global standards.

IMPORTANT NOTICE: *The project is evaluated against Kenya Data Protection Act 2019, ISO/IEC 27001, and PCI DSS. Compliance certifications are claimed only where scope and independent evidence support them.*

Control Domain	Proposed Technical Implementation	Compliance Evidence
Identity & Access	Role-based access, MFA for privileged users, session controls and account lifecycle.	Access review report.
Encryption	TLS for data in transit; encryption at rest and managed keys where supported.	Configuration review.
Secure Development	Code review, dependency scanning, secrets scanning, input validation and security tests.	Build pipeline evidence.
Monitoring	Centralised application/security logs, alerting, time synchronisation and incident runbooks.	Alert and incident drill.
Resilience	Backups, restore testing, recovery targets and controlled failover.	Restore test report.
Independent Assurance	Third-party penetration test and remediation before production approval.	Report and closure evidence.
Data Protection	Data mapping, lawful basis review, privacy notices, retention and DPIA assessment.	Signed privacy decision record.

17. Hosting, Availability, Disaster Recovery and Support

Hosting design should be selected after confirming government hosting policy, data residency needs, traffic assumptions, recovery requirements and total cost.

IMPORTANT NOTICE: *Service level framework includes defined response targets for Critical incidents, High incidents, Routine requests, and Planned maintenance.*

Operational Area	Recommended Technical Approach	Decision to Confirm
Hosting	Approved cloud or data-centre environment with monitored resources and separated environments.	Hosting policy, region and recurring cost.
Backups	Encrypted scheduled backups, separate credentials and periodic restore tests.	Retention and recovery point objective (RPO).
Recovery	Documented recovery runbook, named incident roles and rehearsed restore procedure.	Recovery time objective (RTO) and priorities.
Offline Entry	Approved cached validation or controlled manual fallback, with post-event reconciliation.	Device support and security risk.
Monitoring	Application, infrastructure, payment callback, queue and device-health alerts.	On-call coverage and escalation route.
Service Desk	Ticketed support, severity levels, known-error database and event-day escalation.	Support hours, SLA and service credits.

18. Implementation Roadmap and Duration by Part

The project is structured across seven phases spanning an estimated 10-month delivery timeline.

> ***IMPORTANT NOTICE:*** *Indicative elapsed duration is approximately 10 months. Schedule contingency is required for procurement, device lead times, and venue readiness.*

Phase	Duration	Key Activities & Technical Work	Gate Deliverable
1. Discovery	4 weeks	Workshops, site survey, integrations baseline, risk & requirements.	Signed requirements baseline.
2. Design	4 weeks	UX prototypes, architecture, data model, threat model, DPIA.	Design and privacy gate.
3. Core Build	12 weeks	Website, event catalogue, orders, ticket wallet, admin & reports.	Staging release.
4. Integrations	8 weeks	Payment sandbox, approved scanners, notification & data exchange.	Interface test evidence.
5. Pilot / QA	6 weeks	User acceptance, accessibility, load, security, recovery & simulation.	Pilot acceptance / remediation.
6. Deployment	4 weeks	Production setup, training, operational readiness & launch support.	Go-live approval.
7. Hypercare	4 weeks	Enhanced monitoring, issue closure, handover & service review.	Signed handover certificate.

19. Milestones, Dependencies and Acceptance Gates

Milestones, Dependencies and Acceptance Gates:

> ***IMPORTANT NOTICE:*** *Critical dependencies include tender contract execution, venue site access, payment provider onboarding, hardware vendor cooperation, and safety review sign-offs.*

Target Point	Milestone / Deliverable	Client Dependency	Acceptance Evidence
End Month 1	Discovery report, user journeys, existing inventory & requirements.	Stakeholder access & available drawings.	Signed scope baseline.
End Month 2	Architecture, UX, data model, risk register, privacy decision.	Legal, ICT, ops & data protection review.	Approved design pack.
End Month 4	Core website, admin, ticket lifecycle & reporting in staging.	Fixture rules, pricing & content approvals.	Demo & test results.
End Month 6	Payment sandbox & device integrations; workforce beta.	Provider credentials & vendor cooperation.	Interface test report.
End Month 7	Pilot simulation, load/security/accessibility tests & training draft.	Pilot users & representative test env.	Defect log & risk sign-off.
End Month 9	Production readiness, support rota, recovery & rollback rehearsal.	Production access & formal go-live window.	Readiness checklist signed.
Month 10+	Go-live support, documentation, knowledge transfer & review.	Named operators & support leads.	Handover certificate.

20. Testing, Quality Assurance and Go-Live Readiness

A multi-tiered testing strategy guarantees platform reliability, speed, and security prior to public release.

IMPORTANT NOTICE: *Go-live conditions require zero unresolved critical security issues, verified payment reconciliation, accepted operational runbooks, and formal venue operator sign-off.*

Test Stream	Scope of Testing	Exit Criteria / Evidence
Functional	Ticket lifecycle, pricing, refunds, user roles, event setup, edge cases.	Passed agreed acceptance cases.
Payment	Pending/failed/success callbacks, duplicate callbacks, timeouts, reconciliation.	Provider sandbox evidence & zero discrepancy.
Load / Perf	Peak ticket sales, concurrent sessions, gate validation load, database contention.	Results against agreed workload profile.
Security	Authentication, authorisation, injection, dependencies, secrets, pen testing.	Remediation of Critical/High findings.
Accessibility	Keyboard, screen reader, contrast, focus order, error messages, mobile layout.	Accessibility report & fix log.
Recovery	Backup restore, outage simulation, offline procedure, data reconciliation.	Successful recovery drill.
Operational	Staff roles, escalation, event simulation, support desk, manual exception workflow.	Signed operational readiness review.

21. Project Team, Responsibilities and Knowledge Transfer

Supplier Team Roles & Knowledge Transfer Deliverables:

> ***IMPORTANT NOTICE:*** *Essential client-side roles include Product Owner, Operations Lead, Finance Lead, ICT/Security Lead, DPO, and Procurement Representative.*

Role	Indicative Involvement	Primary Governance Responsibility
Project Manager	Across all phases	Schedule, risks, decisions, status and change control.
Business Analyst	Discovery and acceptance	Requirements, workflows, user journeys and KPIs.
Solution Architect	Design through handover	Architecture, interfaces, quality and technical decisions.
Full-Stack Engineers	Build and stabilisation	Portal, APIs, data model and integrations.
QA Engineer	Build through acceptance	Test plans, regression, performance and acceptance evidence.
Security / Privacy	Design gates and testing	Threat model, privacy review, security tests and remediation.
DevOps Engineer	Environments through support	Deployment, monitoring, backup, recovery and release controls.
Training / Support Lead	Pilot through handover	Training, user manuals, helpdesk and knowledge transfer.

22. Detailed Indicative Budget and Cost Basis

The supplied proposal's KES 75,000,000 total is retained as a preliminary planning envelope for continuity. It has not been independently costed against a final bill of quantities, technical survey, vendor quotations, taxes or tender terms. It must not be represented as a binding price until the scope and exclusions are confirmed.

IMPORTANT NOTICE: *The table does not confirm that turnstiles, cameras, switches, cabling, civil works, SMS fees, payment gateway fees, taxes, or third-party licences are included. The final offer must explicitly state each inclusion and exclusion.*

No.	Work Package / Deliverable Module	Indicative KES	Cost Basis / Inclusions Note
1	**Public ticketing website and mobile web**	**18,500,000**	UX, booking workflow, admin, ticket wallet and payment interfaces.
2	**Access validation and identity integration**	**14,000,000**	Software and approved interfaces; hardware qty to confirm.
3	**Staff / contractor accreditation**	**8,500,000**	Profiles, roles, roster, attendance, expiry and reporting.
4	**Command dashboard and audit analytics**	**7,000,000**	Operational, gate, finance and exception dashboards.
5	**Cybersecurity and privacy assurance**	**4,500,000**	Assessment, test support and remediation allowance.
6	**Device middleware and system interfaces**	**9,500,000**	Depends on device models and interface documentation.
7	**Training, documentation and knowledge transfer**	**3,500,000**	Training sessions, manuals, administrator handover.
SUB	**Indicative Engineering Subtotal**	**65,500,000**	**Engineering, Software & Delivery Subtotal**
8	**First-year hosting / support allowance**	**9,500,000**	Cloud, monitoring, patching and event support assumptions.
TOT	**Indicative Total Planning Envelope**	**75,000,000**	**Preliminary Planning Envelope (Non-binding estimate)**

23. Commercial Model, Payment Milestones and Total Cost of Ownership

A government-facing commercial offer separates one-off delivery costs from recurring service costs.

IMPORTANT NOTICE: *Total Cost of Ownership (TCO) items to price separately include annual hosting, payment provider transaction fees, SMS delivery, hardware maintenance, security re-testing, and event-day support coverage.*

Payment Milestone Trigger	Fee Share %	Acceptance Verification Deliverable
Mobilisation / Approved Inception	**10%**	Contract, mobilisation plan, governance & approved inception pack.
Requirements & Design Approval	**15%**	Signed requirements, architecture, risk & privacy decision pack.
Core Platform Staging Release	**25%**	Demonstrable core modules and agreed staging test evidence.
Integration & Pilot Acceptance	**20%**	Integration tests, pilot report and agreed remediation status.
Production Readiness / Deployment	**20%**	Go-live checklist, security sign-off and operational rehearsal.
Handover & Documentation	**10%**	Signed training, documentation, handover and support transition.

24. Benefits Measurement, Risk Register and Mitigation

Risk Register and Mitigation Matrix:

IMPORTANT NOTICE: *Monthly benefit measurement tracks financial reconciliation, checkout completion rate, median validation speed, system uptime, and roster compliance.*

Risk Description	Likelihood / Impact	Proposed Mitigation Strategy	Owner
Unclear existing interfaces	Medium / High	Site survey, vendor workshops, interface proof-of-concept, explicit exclusions.	Client ICT / Lead
Privacy / biometric challenge	Medium / High	Necessity test, legal review, DPIA where required, alternative entry.	Client DPO / Legal
Payment callback mismatch	Medium / High	Sandbox tests, idempotency, provider status checks, daily reconciliation.	Finance Lead
Connectivity / power outage	Medium / High	Tested offline procedure, power/network monitoring, recovery drill.	Venue ICT / Ops
Scope growth / cost overrun	High / High	Baseline scope, change requests, assumptions register, itemised pricing.	Steering Committee
User adoption issues	Medium / Medium	Usability tests, clear signage, staff training, support desk.	Service Lead
Vendor lock-in	Medium / Medium	Open interfaces, data export format, documentation, exit plan.	Client Procurement
Schedule conflict with events	Medium / High	Critical-path plan, event calendar, pilot windows, go/no-go gates.	Project Manager

25. Governance, Acceptance, Sources and Final Submission Checklist

Governance Accountabilities & Final Submission Checklist:

- Confirm actual recipient, tender reference, submission format, deadline and mandatory forms.
- Verify current venue handover status and confirm site access and system inventory.
- Obtain vendor quotes and validate taxes, hardware, licences, hosting, connectivity and support assumptions.
- Attach genuine company registration, tax compliance, experience references, CVs and certifications.
- Obtain qualified technical, legal, data-protection, cybersecurity and procurement review.
- Verified public context sources: CRBC update dated 10 Feb 2026; Presidential inspection report dated 11 April 2025.

Stakeholder Group	Primary Role & Governance Accountability
Government / Venue Sponsor	Scope, funding pathway, major decisions and formal acceptance authority.
Venue Operations	Matchday workflows, safety coordination, gate procedures and operational acceptance.
ICT / Cybersecurity	Architecture, hosting, interfaces, security assurance and continuity approval.
Data Protection / Legal	Lawful basis, privacy notices, DPIA decision, retention and biometric review.
Finance / Internal Audit	Reconciliation definitions, reporting and audit access.
ST-Tech ICT Consultants (Supplier)	Deliver, test, document, train and support approved scope under contract.

26. Detailed Public Website Sitemap and User Journeys

This annex turns the public-facing concept into a buildable information structure.

Page / Area	Key Functional Features	Acceptance Verification Checks
Home Page	Official notices, fixtures, venue guide, ticket call-to-action, support status.	Mobile layout, accessible headings, approved content.
Events Page	Search/filter events, date pickers, event status, ticket availability.	Correct timezone, closed/sold-out states.
Event Details	Venue map, entry rules, category pricing, refund terms, accessibility.	Terms visible before payment.
Checkout	Order summary, contact details, provider handoff, payment status, retry.	No duplicate order or false paid status.
Ticket Wallet	Confirmed ticket, reference, resend/recovery, support link.	Ticket privacy and revocation behaviour.
Help / Policies	FAQ, refunds, privacy notice, fraud warning, accessibility support.	Contacts and policies approved.

27. Seat Mapping, Ticket Categories and Inventory Rules

Seat maps shown are conceptual zones, not a surveyed seating plan. Production inventory must reflect approved seating charts.

Inventory Rules to Implement:

- Create event-specific inventory; never assume every physical seat is saleable for every event.
- Represent seats, blocks, standing zones, accessible allocations, restricted seats and sponsor holds separately.
- Use transactional locking or equivalent concurrency control to prevent overselling under high traffic.
- Apply temporary holds only for a defined time and release expired holds automatically.
- Record every manual allocation, price change, refund, complimentary ticket and inventory override in an audit trail.
- Support staged releases of ticket allocations where approved, without exposing restricted inventory.
- Offer a clear seat/zone description and accessible alternative when a section is unavailable.

IMPORTANT NOTICE: *High-impact changes to event sale windows or pricing require dual-approver administrative authorization.*

28. API Design, Integration Contracts and Error Handling

API Design, Integration Contracts and Error Handling Specifications:

> ***IMPORTANT NOTICE:*** *Failure patterns planned for include duplicate provider callbacks, slow payment gateway retries, offline scanner operation, and degraded external services.*

Interface Endpoint	Primary Purpose	Essential Security Safeguards
Event API	Read published events, categories, sale windows and venue notices.	Role-based writes, validation, public caching.
Order API	Create order, reserve inventory, query order status, cancel draft.	Idempotency keys, expiry and concurrency locks.
Payment Adapter	Initiate approved provider flow and receive status callbacks.	Signature check, authentication, retries, reconciliation.
Validation API	Check ticket validity and record gate entry result.	Replay protection, device auth, rate limits.
Workforce API	Read approved roster and credential permissions.	Minimal data, permission checks and expiry.
Reporting API	Provide approved event, finance and operations reports.	Export authorisation, filtering and audit trail.

29. Database, Audit Trail and Data Lifecycle

Database Architecture, Audit Trail and Data Lifecycle:

IMPORTANT NOTICE: *Database operational controls require tracked schema migrations, encrypted backups, separation of prod/dev environments, and transactional inventory locks.*

Data Group	Examples	Lifecycle & Protection Standard
Operational	Events, ticket categories, orders, validation records, staff assignments.	Retention by approved operational and legal schedule.
Payment References	Provider transaction IDs, amount, currency, state, outcome.	Do not store unnecessary payment credentials.
Identity / Contact	Account identifier, contact details and preferences.	Purpose limitation, access controls, deletion process.
Security Audit	Administrator action, actor, timestamp, object, result.	Restricted write access, monitored integrity, retention.
Sensitive / Biometric	Only if separately approved and strictly necessary.	Separate legal gate, minimal retention, access logging.

30. Hosting, DevSecOps and Deployment Controls

Proposed Deployment Workflow:

- Commit code to source control with mandatory peer review and branch protections.
- Run automated unit, integration, lint, dependency and secret-scanning checks in CI/CD pipeline.
- Build versioned release container artefact and record provenance.
- Deploy to staging; run automated acceptance tests against representative dataset.
- Approve production change with rollback and database migration plan.
- Deploy gradually where possible; verify health checks, payment callbacks and ticket validation.
- Monitor errors and latency; rollback or disable feature if anomalies detected.

Environment Rules: Production credentials must be stored in a managed secret vault, never in source code. Development, staging, training, and production environments are logically isolated.

31. Event-Day Operations and Command-Centre Runbook

Event-Day Operations and Command-Centre Runbook:

> ***IMPORTANT NOTICE:*** *The platform supports the event command structure; it does not replace emergency plans, safety officers, police, medical teams or venue life-safety systems.*

Timing Phase	Operational Action Required	Responsible Owner / Evidence
Before Event	Confirm event config, categories, approved roster, scanner health, network/power, support contacts.	Event lead signs readiness checklist.
Gate Opening	Verify sample tickets, watch rejects and device heartbeat, confirm supervisor escalation route.	Gate supervisor records checks.
During Event	Monitor queues, incidents, system health, payment and customer support escalations.	Command lead tracks incidents.
After Event	Close sales as approved, export attendance/finance reports, reconcile exceptions, secure devices.	Operations and finance sign closeout.
Post-Event Review	Review defects, customer complaints, gate throughput, outages, and corrective action log.	Lessons-learned action log.

32. Load Planning, Performance Targets and Capacity Validation

Capacity must be engineered from an agreed demand model, not headline stadium seating numbers.

IMPORTANT NOTICE: *Discovery phase targets to agree include peak sales load profiles, max acceptable API response latency, RTO/RPO objectives, and monitoring alert thresholds.*

Workload Stream	What to Model & Stress Test	Evidence Required
Ticket Release	Concurrent sessions, seat/zone contention, cart holds and checkout conversion.	Load-test report with assumptions.
Payment Callbacks	Provider latency, callback retries, status queries and queue backlog.	Duplicate/delayed callback scenarios.
Gate Validation	Peak scan rate per lane, offline behaviour and sync after outage.	Representative device and network test.
Reports & Dashboards	Concurrent dashboard refreshes, exports and large event history queries.	Query performance and access tests.
System Recovery	Restore duration, data recovery point and failover behaviour.	Observed restore/failover exercise.

33. Training, Service Desk and Warranty Framework

Training Streams and Warranty Support Framework:

> ***IMPORTANT NOTICE:*** *Support model includes defined contract hours, ticketed severity escalation, known-error database maintenance, and a 12-month defect warranty.*

Training Stream	Target Audience	Core Topics Covered	Proof of Readiness
Event Administrators	Event Admins	Event setup, categories, sale windows, notices, approvals.	Practical event setup exercise.
Gate Supervisors	Supervisors / Stewards	Scanner workflow, exceptions, offline procedure, escalation.	Simulated gate scenario.
Finance / Audit	Finance Officers	Settlement reconciliation, refunds, exports, audit trail.	Sample event reconciliation signed.
ICT / Security	IT & Security Staff	Monitoring, access reviews, releases, backup/restore, incidents.	Recovery and incident drill.
Helpdesk	Support Agents	Ticket recovery, customer comms, escalation, verification.	Role-played support cases.

34. Procurement Evaluation, Acceptance Matrix and Change Control

Procurement Evaluation Matrix and Acceptance Controls:

> ***IMPORTANT NOTICE:*** *Material change requests must log reason, options, cost/schedule impact, risk impact, and formal client approval before execution.*

Evaluation Area	Evidence Required to Prepare	Responsible Owner
Technical Compliance	Requirement-by-requirement response, architecture, security, accessibility, integrations.	Technical Lead
Delivery Capacity	Named CVs, realistic schedule, project governance and resource plan.	Bid Manager
Relevant Experience	Verifiable references, signed completion certificates, client contacts where permitted.	Authorised Representative
Financial Offer	Itemised pricing, taxes, recurring fees, assumptions, exclusions, payment terms.	Finance Lead
Legal / Eligibility	Registration, tax status, declarations and other documents explicitly requested by tender.	Compliance Lead
Implementation Assurance	Testing plan, privacy decision, support plan, acceptance gates, risk register.	Delivery Lead

35. Final Technical Commitments, Assumptions and Next Actions

What the proposed system is intended to provide:

- An official, mobile-friendly event and ticketing website with clear prices and terms.
- Secure order and payment-state handling with reconciliation and support workflows.
- Ticket validation and authorised staff accreditation, with tested exception procedures.
- Role-based operational, gate-health, workforce and financial dashboards.
- Documented cybersecurity, privacy, backup, incident, release and support controls.
- Training, manuals, source/configuration documentation and a planned handover.
- A modular foundation that can be extended only through approved scope and change control.

Assumptions that must be confirmed:

- The client will identify the authorised sponsor, operator, product owner and acceptance authority.
- Current venue handover, site access, equipment inventory and existing vendor interfaces will be confirmed.
- The client will provide approved event rules, ticket policies, price rules and content.
- Payment providers, messaging providers, hosting arrangements and third-party costs will be selected and priced.
- Any biometric processing remains out of scope unless separately justified, approved, priced and contracted.
- The KES 75 million planning envelope and 10-month timeline are indicative, not binding commitments.

Recommended immediate next steps: 1. Confirm legal proposer and procurement route. 2. Obtain official tender requirements. 3. Run a discovery/site survey. 4. Verify vendor and payment interfaces. 5. Complete technical, legal, privacy and security reviews. 6. Produce a priced bill of quantities and compliance matrix. 7. Submit only verified documents and evidence.

> ***IMPORTANT NOTICE:*** *A successful stadium digital platform is more than a website: it is a tested operating service with accountable ownership, transparent finance, resilient gate operations, and clear governance.*